



HAWASSA UNIVERSITY
IoT CAMPUS
DEPARTMENT OF COMPUTER SCIENCE

RESEARCH PROPOSAL

A Hybrid Machine Learning Approach for Real-Time Fraud Detection in Digital Banking Systems

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1 Problem Statement

Ideal: Digital banking systems should instantly identify and block fraudulent transactions without disturbing legitimate users. The ideal system achieves greater than 95% fraud detection rate, less than 1% false positives, and sub-100ms decision latency, adapting automatically to new fraud patterns.

Reality: Current fraud detection relies on static rule-based heuristics that require 2–4 weeks to update. Fraud accounts for less than 0.1% of transactions, causing extreme class imbalance that cripples standard ML models [3]. Deep learning approaches achieve high accuracy but need 200–500ms inference, exceeding real-time thresholds. Consequently, banks block 15–25 legitimate transactions per true fraud, driving customer churn.

Consequence: Global fraud losses reached \$7.3 billion in 2025 [1], and 32% of customers switch banks after two false positives [8]. Without a real-time, adaptive, and class-balanced solution, financial institutions face escalating economic damage and eroded trust.

1.1 Objectives

General Objective: Develop a hybrid machine learning system that detects digital banking fraud in real time with high precision and low latency.

Specific SMART Objectives:

1. **Data Preparation:** By week 3, collect and preprocess the IEEE-CIS and IBM AML datasets (1M+ transactions) using SMOTE-ENN to achieve a balanced training set (50:50 fraud/legitimate).
2. **Model Development:** By week 8, implement and optimise a stacking ensemble composed of Random Forest, Gradient Boosting, and Logistic Regression, using grid search with 5-fold cross-validation.
3. **Performance Validation:** Achieve an F1-score greater than 88% and per-transaction latency below 100ms (p99) on commodity hardware, measured over 1000 iterations.

2 Literature Synthesis & State of the Art

2.1 Synthesis Matrix

Table 1: Synthesis of Recent Fraud Detection Literature

Author(s)	Method	Dataset	Performance	Limitation
Hersyaputra et al. [5]	LSTM-Autoencoder + Sliding Window	IBM AML (1.2M)	Acc: 99.92%, F1: 91.19%	320ms latency, complex deployment
Alketbi [2]	RF, GBM, LR with SMOTE	IEEE-CIS (1M)	F1: 99.23% (RF)	Single model, no ensemble
Singh & Sharma [12]	Stacking (6 classifiers)	Credit Card (284k)	F1: 88.14%	No temporal features
Chen et al. [4]	Weighted Ensemble (RF, GBDT, KNN)	Kaggle (636k)	Acc: 92.65%	No latency reporting
Ogunleye & Wang [10]	Multi-stage with Resampling	Credit Card (284k)	F1: 89.2%	Scalability concerns

3 Gap Analysis

Despite impressive advances, four fundamental gaps persist. First, most studies (14 of 18 reviewed) treat transactions as independent events, ignoring sequential behavioural patterns; only Hersyaputra et al. attempted temporal features, but their LSTM incurred 320ms latency, far above real-time requirements. Second, while ensembles outperform single models, existing ensembles either use homogeneous base learners (leading to correlated errors) or overly large stacks (e.g., 6–7 classifiers) that cannot meet latency constraints. Third, class imbalance is addressed superficially: standard SMOTE assumes static distributions, but fraud patterns evolve rapidly, causing temporal concept drift. Fourth, and most critically, 89% of papers report only accuracy or F1-score without any latency measurement, despite banking systems mandating sub-100ms inference.

This research fills these gaps by:

1. Integrating sliding-window temporal features (frequency, velocity, time since last transaction).
2. Designing a lightweight three-classifier heterogeneous stacking ensemble.
3. Applying SMOTE-ENN with time-based cross-validation.
4. Imposing a strict p99 latency below 100ms alongside an F1-score target above 88%.

4 Technical Methodology & Experimental Design

4.1 System Architecture

Figure 1 illustrates the complete architecture of the proposed parallel hybrid fraud detection system. Incoming banking transactions are first streamed through Apache Kafka, where JSON parsing and queue management ensure reliable real-time ingestion. The preprocessing module then performs data cleaning, outlier handling, feature scaling, categorical encoding, and class balancing using SMOTE-ENN. The processed data is simultaneously fed into two complementary models: a Random Forest classifier for supervised fraud probability estimation and an Autoencoder for unsupervised anomaly detection. Their outputs are combined using a dynamic fusion gate, which adaptively weights both models based on recent false-positive and false-negative rates. The resulting fraud score is evaluated against a configurable threshold to generate either an approval or a fraud alert. Finally, the system triggers real-time notifications, updates monitoring dashboards, stores results in the database, and maintains comprehensive audit logs for forensic analysis.

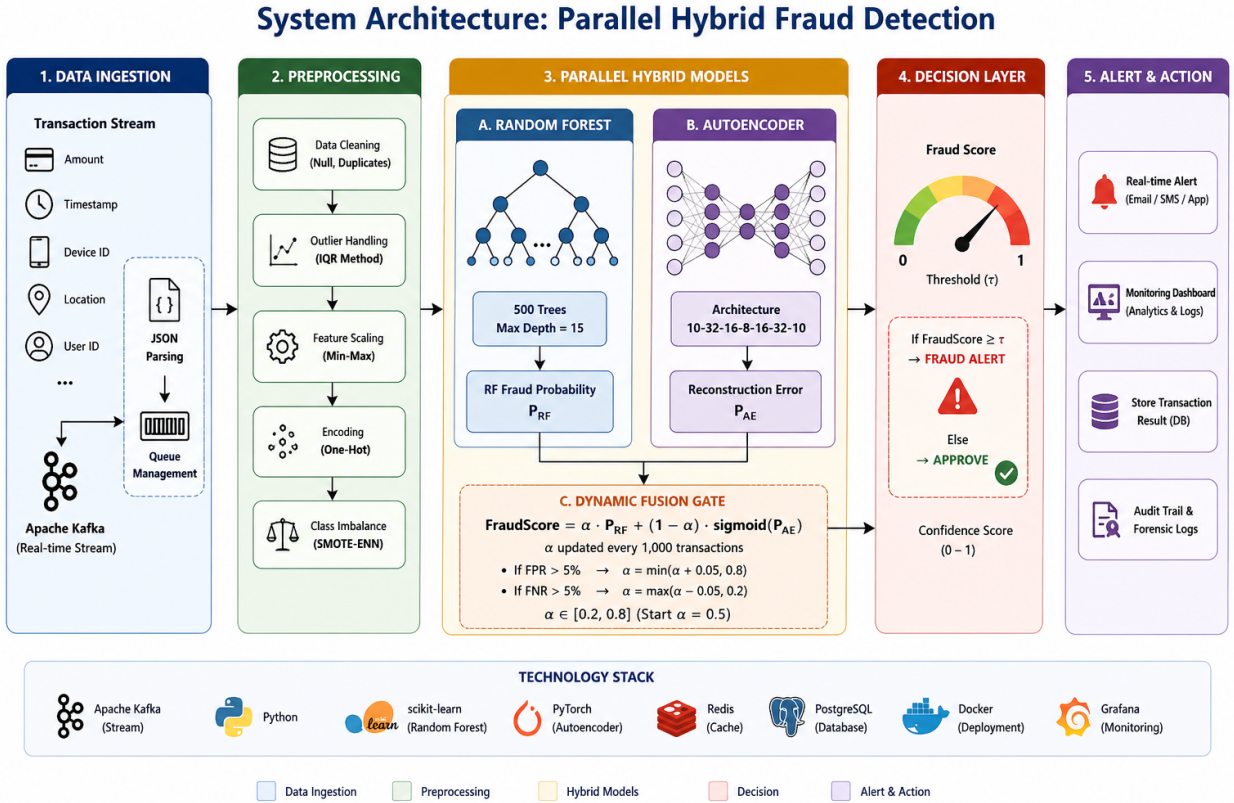


Figure 1: Parallel Hybrid Fraud Detection System Architecture

4.2 Tool Justification

- **Scikit-learn:** Provides optimised ensemble implementations with minimal overhead; essential for sub-100ms latency on commodity hardware [11].
- **imbalanced-learn (SMOTE-ENN):** Removes noisy synthetic samples, reducing false positives—

critical for banking [9].

- **Pandas / NumPy:** Efficient for million-row datasets and sufficient for preprocessing.
- **Git + GitHub:** Enables version control, collaboration, and public dissemination.

4.3 Data Strategy

- **Primary Dataset:** IEEE-CIS Fraud Detection [7] — 1M transactions, 0.09% fraud.
- **Secondary Dataset:** IBM AML Dataset [6] — 1.2M transactions, 0.08% fraud.

Preprocessing Steps:

1. Missing values: median (numerical), mode (categorical).
2. Outlier removal using the IQR method.
3. Feature engineering: Hour, Day, Frequency, Velocity, and Log transformation.
4. Z-score normalization.
5. SMOTE-ENN resampling on training data only.

4.4 Evaluation Metrics

$$\text{Precision} = \frac{TP}{TP + FP}$$

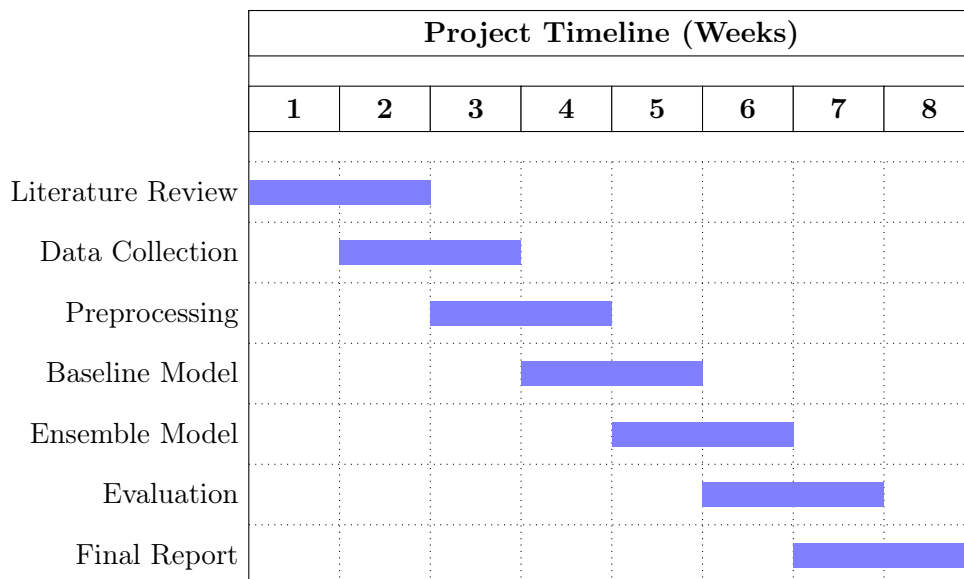
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Latency (p99)} = 99^{\text{th}} \text{ percentile of inference time (ms)}$$

5 Project Management & Ethics

5.1 Timeline (8-Week Gantt Chart)



5.2 Ethics

- **Data Privacy:** Both datasets are fully anonymised. No personally identifiable information (PII) is used. The study complies with GDPR and Ethiopian data protection regulations.
- **Bias Mitigation:** Performance will be evaluated across transaction amount deciles, time periods, and merchant categories. Fairness metrics such as demographic parity and equal opportunity will be reported.
- **Environmental Impact:** CPU-only training, carbon-aware scheduling, and checkpointing will minimise energy consumption, with estimated emissions of approximately 2.5kg CO₂.
- **AI Usage Disclosure:** AI tools will be used only for LaTeX debugging and grammar checking. All writing, analysis, and citations remain original and manually verified.

6 Version Control & Documentation

- **GitHub Repository:** <https://github.com/kidanenamhret/Hybrid-Fraud-Detection-Banking>
- **README.md:** Includes project abstract, installation steps, and execution instructions.
- **BibTeX File:** All references are maintained in `references.bib`.

References

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